

From Holographic Space-Time to The Triadic Cosmos

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Abstract

The Triadic Cosmos framework consolidates several recent architectural insights about the holographic nature of the universe, the role of electromagnetic time, and the emergence of bulk geometry from boundary information. In this view, the universe is fundamentally holographic: all physical information resides on a two-dimensional cosmic boundary, while the three-dimensional bulk arises as a decode of this boundary data. Electromagnetic time is treated as a universal and fundamental temporal parameter with Planck-scale resolution, governing both vacuum phenomena and the propagation of gravitational and electromagnetic degrees of freedom. By contrast, material time is emergent and closure-specific, generated through a triadic recursion cycle linking energy, geometry, and information. Closures form a fractal hierarchy ranging from black holes as minimal horizon-bounded domains to the cosmic boundary as the maximal closure.

This paper positions Triadic Cosmos within the broader holographic tradition and highlights its compatibility with mainstream frameworks that adopt similar architectural assumptions. In particular, we examine Holographic Space-Time (HST) as a representative theory in which time is fundamental, geometry is emergent, and horizon-bounded causal domains encode physical information. We show that Triadic Cosmos extends these ideas by providing a dynamical dual-time structure, a fractal closure architecture, and a unified triadic cycle that governs internal evolution. We further outline how the framework aligns with other holographic and emergent approaches, including AdS/CFT, tensor-network geometry, causal set theory, emergent gravity, and black-hole thermodynamics. Triadic Cosmos thus offers a coherent architectural completion of the holographic paradigm while remaining fully compatible with established theoretical directions.

1 Introduction

Modern theoretical physics increasingly converges on a set of architectural principles that transcend individual formalisms: holography as the organizing structure of physical information, emergent geometry as the manifestation of that information in the bulk, and a fundamental temporal parameter governing electromagnetic and gravitational propagation. The *Triadic Cosmos* framework consolidates these principles into a unified architectural ontology, integrating recent insights regarding the role of the vacuum, the status of electromagnetic time, and the emergence of material time within horizon-bounded domains.

In this framework, the universe is fundamentally holographic. All physically meaningful information resides on a two-dimensional cosmic boundary, while the three-dimensional bulk arises as a decode of this boundary data. Spatial geometry is therefore not fundamental but emergent: it reflects the informational structure and entanglement patterns encoded at the boundary. Gravity, in turn, is interpreted as a geometric effect of boundary information rather than as a fundamental interaction.

A central insight of Triadic Cosmos is that electromagnetic time is *fundamental*. Vacuum phenomena—virtual excitations, Casimir effects, horizon redshift—occur entirely in electromagnetic time, and the vacuum itself possesses no horizon, no informational capacity, and no recursion index. Because the vacuum is not a closure, it cannot host emergent temporal flow; thus electromagnetic time must serve as the universal temporal parameter of the boundary, with Planck-time as its minimal resolution.

Material time, by contrast, is emergent and closure-specific. Physical reality is organized into a fractal hierarchy of closures: horizon-bounded domains with finite informational capacity. Black holes represent minimal closures, observers and physical systems form intermediate closures, and the cosmic boundary constitutes the maximal closure. Within each closure, internal evolution is governed by a triadic recursion cycle linking energy, geometry, and information. This cycle,

$$E \rightarrow G \rightarrow I \rightarrow E,$$

generates material time, whose rate is accelerated by entropy and damped by mass. Black-hole evaporation provides the empirical calibration for this dual-time structure, yielding the scaling relation

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2}.$$

Having established the architectural foundations of Triadic Cosmos, it is natural to examine its relationship to existing mainstream frameworks that adopt similar assumptions. Among these, *Holographic Space-Time* (HST), developed by Banks and collaborators, is a particularly relevant representative. HST treats time as fundamental, geometry as emergent, and horizon-bounded causal diamonds as the basic informational units of physical reality. These principles align closely with the Triadic Cosmos architecture. At the same time, Triadic Cosmos extends this shared foundation by introducing a dynamical dual-time structure, a fractal closure hierarchy, a unified triadic recursion cycle, and a vacuum interpretation consistent with holographic projection.

The final part of this paper situates Triadic Cosmos within the broader holographic tradition, highlighting its compatibility with other mainstream approaches such as AdS/CFT, tensor-network emergent geometry, causal set theory, emergent gravity, and black-hole thermodynamics. Triadic Cosmos thus provides a coherent architectural completion of the holographic paradigm while remaining fully aligned with established theoretical directions.

2 The Triadic Cosmos Architecture

The Triadic Cosmos framework is built upon a minimal set of architectural principles that jointly define the structure of physical reality. These principles integrate holography, emergent geometry, dual time, fractal closure organization, and a universal dynamical cycle linking energy, geometry, and information. In this section we present the core components of the architecture in a systematic manner.

2.1 Holographic Universe

The universe is fundamentally holographic. All physically relevant information is encoded on a two-dimensional cosmic boundary, while the three-dimensional bulk arises as a decode of this boundary data. Bulk geometry, matter fields, and causal structure are emergent manifestations of boundary information. This holographic organization ensures that the boundary serves as the primary informational substrate of the cosmos.

2.2 Emergent Geometry

Spatial geometry in the bulk is not fundamental. Instead, it emerges from the informational structure and entanglement patterns encoded at the boundary. The metric properties of the bulk reflect constraints imposed by holographic decoding, and gravitational phenomena are interpreted as geometric responses to boundary information. In this view, gravity is not a fundamental interaction but an emergent geometric effect.

2.3 Fundamental Electromagnetic Time

A central architectural insight is that electromagnetic time (t_{EM}) is fundamental. It constitutes the universal temporal parameter of the boundary and possesses Planck-scale resolution. Vacuum phenomena—virtual excitations, Casimir effects, horizon redshift—occur entirely in t_{EM} . Because the vacuum is not a closure and therefore lacks horizon structure, informational capacity, and recursion, it cannot host emergent temporal flow. Consequently, electromagnetic time must be taken as fundamental.

2.4 Fractal Closure Hierarchy

Physical reality is organized into closures: horizon-bounded domains with finite informational capacity. Closures form a fractal hierarchy:

- minimal closures: black holes,
- intermediate closures: observers, physical systems, causal domains,
- maximal closure: the cosmic boundary.

Each closure possesses its own internal informational structure and supports emergent material time. The universe itself is treated as a closure of highest order, providing a global boundary for all physical processes.

2.5 Dual Time Structure

Triadic Cosmos introduces a dual-time architecture:

1. **Electromagnetic time** (t_{EM}): fundamental, boundary-defined, universal.
2. **Material time** (t_{mat}): emergent, closure-specific, generated by internal recursion.

Material time accelerates with entropy and is damped by mass. Black-hole evaporation provides the empirical calibration for this dual-time structure:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2}.$$

This relation reflects the dependence of closure-internal temporal flow on horizon area and informational capacity.

2.6 Triadic Recursion Cycle

The dynamical engine of the Triadic Cosmos is a triadic recursion cycle linking energy (E), geometry (G), and information (I):

$$E \rightarrow G \rightarrow I \rightarrow E.$$

This closed cycle governs internal evolution within closures and generates material time. Energy drives geometric configuration, geometry constrains informational structure, and information feeds back into energetic degrees of freedom. The triadic cycle provides the missing dynamical layer absent from purely kinematical holographic frameworks.

2.7 Vacuum as an EM-Time Medium

The vacuum is not a closure. It has no horizon, no informational capacity, and no recursion index. As a result, it supports no material time and no triadic evolution. Instead, the vacuum functions as a pure electromagnetic-time medium in which holographic decoding manifests as bulk geometry and matter. This interpretation resolves inconsistencies in quantum field theory and cosmology regarding vacuum dynamics and temporal behavior.

2.8 String Theory as Holographic Encoding Layer

String theory is incorporated as the encoding layer of the holographic boundary. The graviton is interpreted as a boundary operator, and Calabi–Yau structures serve as informational substrates for holographic encoding. Because the vacuum is not dynamical, the framework remains compatible with string theory while avoiding issues associated with dynamical vacua.

Together, these architectural components define the Triadic Cosmos as a coherent, structurally complete holographic ontology. In the following section, we examine how this architecture relates to Holographic Space–Time and other mainstream frameworks that share similar foundational assumptions.

2.9 Global Hilbert–Space Ontology

A foundational architectural principle of the Triadic Cosmos is that the universe is described by a single global Hilbert space,

$$\mathcal{H}_{\text{universe}},$$

within which all physically meaningful states reside. Closures correspond to nested finite–dimensional subspaces,

$$\mathcal{H}_{\text{closure}} \subset \mathcal{H}_{\text{universe}},$$

reflecting the finite informational capacity imposed by holographic area bounds. This Hilbert–space ontology ensures that energetic, geometric, and informational roles are consistently embedded across all scales, and that emergent bulk geometry arises from boundary degrees of freedom rather than from fundamental spatial fields.

The global Hilbert space provides the structural foundation for holography, dual time, and fractal closure organization. It also guarantees that physical reality is constrained by finite informational capacity, enabling the triadic recursion cycle to operate coherently within each closure.

2.10 Planck Length as Holographic Resolution Bound

Although spatial geometry is emergent and formally continuous, holography imposes a finite informational capacity on any horizon,

$$S = \frac{A}{4G\hbar},$$

implying a maximal number of independent degrees of freedom,

$$N_{\text{states}} \sim \frac{A}{\ell_P^2}.$$

For a spherical closure, the relation between volume and area yields an effective minimal bulk resolution scale of order ℓ_P . This scale is not a geometric pixel size but an information–theoretic bound on distinguishability: two bulk configurations cannot be holographically distinguished below the Planck scale.

Thus, the Planck length arises naturally as the minimal holographic resolution of emergent geometry, reflecting the finite dimensionality of closure Hilbert spaces rather than any fundamental discretization of spacetime.

2.11 Triadic Filters as Structural Constraints

The triadic architecture imposes four structural filters that determine which geometric configurations can correspond to physical reality. These filters arise from the requirement that the triadic recursion cycle,

$$E \rightarrow G \rightarrow I \rightarrow E,$$

must be well-defined within each closure.

1. **Filter 1: Absence of a Required Role.** Structures lacking energy, geometry, or information cannot support recursion. This excludes white holes, naked singularities, tachyonic geometries, and information-free horizons.
2. **Filter 2: Negative Roles.** Negative energy, negative geometry, or negative information violate holographic and thermodynamic constraints. This excludes warp drives, traversable wormholes, Tipler cylinders, and Gödel-type retrocausal geometries.
3. **Filter 3: Inconsistent Information Embedding.** Geometries with discontinuous or non-holographic informational structure cannot support the update $G_n \rightarrow I_{n+1}$. This excludes exotic compact objects such as gravastars, boson stars, fuzzballs, and dark-energy stars.
4. **Filter 4: Failure of Triadic Closure.** If the full cycle $E \rightarrow G \rightarrow I \rightarrow E$ cannot close, recursion and emergent time become undefined. This excludes all geometries that cannot reorganize energetic, geometric, and informational roles across successive states.

These filters identify the subset of GR geometries that correspond to physical reality and exclude mathematically valid but architecturally incomplete solutions. They provide the structural mechanism through which the Triadic Cosmos reconciles GR with holography, quantum field theory, and the Standard Model.

2.12 General Relativity within the Triadic Architecture

General Relativity (GR) remains fully valid within the Triadic Cosmos, but its mathematical structure is architecturally broader than the subset of geometries that can be physically realized in a finite-capacity holographic universe. GR is formulated on smooth manifolds with continuous metric fields, intentionally allowing a vast solution space that includes non-physical configurations such as white holes, naked singularities, traversable wormholes, and Gödel-type universes. From the perspective of the Triadic Cosmos, GR is therefore *overcomplete*: it permits more geometric degrees of freedom than can be instantiated within closures possessing finite informational capacity and well-defined triadic recursion.

Within the triadic architecture, physically meaningful geometries must satisfy holographic capacity bounds, support consistent informational embedding, and admit full triadic closure. These requirements restrict GR to the subset of geometries that can be represented within finite-dimensional closure Hilbert spaces. The continuity of GR is thus interpreted as a macroscopic limit of an underlying discrete state structure: geometry is emergent and continuous as a decode, while the space of physically distinguishable states is discrete due to finite boundary information.

This motivates the global Hilbert-space ontology adopted by the Triadic Cosmos. The universe is described by a single Hilbert space,

$$\mathcal{H}_{\text{universe}},$$

with closures represented as nested finite-dimensional subspaces,

$$\mathcal{H}_{\text{closure}} \subset \mathcal{H}_{\text{universe}}.$$

This ontology is compatible with holography, emergent geometry, dual time, and fractal closure hierarchy. Alternative ontologies—classical phase space, timeless constraint spaces, or locally defined Hilbert spaces—either lack coherent holographic encoding, fail to support nested closure structure, or do not provide a universal temporal parameter compatible with electromagnetic time.

In this architectural context, GR retains its full descriptive power for physically realizable geometries, while mathematically valid yet triadically incomplete solutions are excluded. GR provides the geometric language; the Triadic Cosmos supplies the structural constraints that determine which GR geometries correspond to physical reality.

2.13 Recursion as Holographic Renormalization Flow

The recursive structure of the Triadic Cosmos admits a natural holographic interpretation. In holographic dualities such as AdS/CFT, the radial direction of the bulk spacetime plays the role of a renormalization scale: moving inward corresponds to coarse-graining the boundary theory, while moving outward corresponds to refining it. This structure is inherently recursive, with each radial step reorganizing the energetic, geometric, and informational degrees of freedom of the boundary.

The triadic recursion cycle,

$$X_{n+1} = R(X_n),$$

can therefore be interpreted as a discrete analogue of holographic renormalization flow. The recursion index n corresponds to a coarse-graining scale, and each triadic update reorganizes the structural roles according to their mutual constraints. Under this identification:

- the transition $E \rightarrow G$ reflects how boundary energetic content shapes bulk geometry under RG flow,
- the transition $G \rightarrow I$ reflects how geometric structure bounds informational capacity (e.g. holographic area laws),
- the transition $I \rightarrow E$ reflects how informational structure constrains energetic evolution (e.g. unitarity, entropy bounds).

Fixed points of the triadic recursion correspond to conformal fixed points of holographic renormalization, where the structural roles mutually stabilize. This mapping clarifies why holographic frameworks satisfy the architectural constraints of the Triadic Cosmos and shows that triadic recursion provides a minimal architectural mechanism underlying holographic RG flow and emergent temporal structure.

3 Relation to Holographic Space–Time

Holographic Space–Time (HST), developed by Banks and collaborators, is one of the few mainstream frameworks that explicitly adopts the architectural principles of fundamental time, emergent geometry, and holographic organization. Although formulated independently, HST shares several structural assumptions with the Triadic Cosmos and therefore provides a useful reference point for situating the present framework within the broader holographic tradition. In this section we outline the points of alignment and the architectural extensions introduced by Triadic Cosmos.

3.1 Shared Architectural Principles

HST is built upon three core assumptions that closely parallel those of the Triadic Cosmos:

1. **Fundamental Time.** HST treats time as a fundamental evolution parameter associated with the boundary Hilbert space of each causal diamond. This aligns with the Triadic Cosmos identification of electromagnetic time as the universal and fundamental temporal parameter.
2. **Emergent Geometry.** In HST, bulk geometry is not fundamental but reconstructed from the holographic degrees of freedom associated with horizon areas. This matches the Triadic Cosmos view in which spatial geometry arises from boundary information and holographic decoding.

3. **Horizon–Bounded Domains.** HST models physical reality through causal diamonds—finite, horizon–bounded regions with well–defined informational capacity. These structures correspond directly to the closure hierarchy of the Triadic Cosmos, where black holes represent minimal closures and the cosmic boundary represents the maximal closure.

These shared principles establish a strong architectural compatibility between the two frameworks. Both treat holography as fundamental, geometry as emergent, and horizon structure as the organizing principle of physical information.

3.2 Architectural Extensions Provided by Triadic Cosmos

While HST provides a powerful kinematical formulation of holographic spacetime, it does not explicitly supply the dynamical, temporal, and fractal layers required for a complete architectural description. The Triadic Cosmos extends HST along several key dimensions:

1. **Dual Time Structure.** HST employs a single fundamental time parameter. Triadic Cosmos introduces a dual–time architecture in which electromagnetic time is fundamental and material time is emergent, closure–specific, and generated by triadic recursion. This provides a dynamical mechanism for temporal variation across closures.
2. **Fractal Closure Hierarchy.** HST models causal diamonds but does not formulate a fractal hierarchy of closures. Triadic Cosmos generalizes horizon–bounded domains into a nested, fractal structure ranging from black holes to the cosmic boundary, thereby providing a unified organizational ontology.
3. **Triadic Recursion Cycle.** HST lacks an explicit dynamical engine linking energy, geometry, and information. Triadic Cosmos introduces the triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow E,$$

which governs internal evolution within closures and generates material time.

4. **Vacuum Architecture.** HST does not provide a structural interpretation of the vacuum. Triadic Cosmos treats the vacuum as an electromagnetic–time medium without closure, informational capacity, or recursion, resolving inconsistencies in temporal behavior and holographic projection.
5. **Universe as Maximal Closure.** HST does not explicitly define a global closure corresponding to the entire universe. Triadic Cosmos identifies the cosmic boundary as the maximal closure, completing the fractal hierarchy and providing a global holographic substrate.
6. **String–Theoretic Encoding Layer.** While HST is compatible with string theory, it does not specify an encoding mechanism. Triadic Cosmos incorporates string theory as the holographic encoding layer, interpreting gravitons as boundary operators and Calabi–Yau structures as informational substrates.

These extensions transform the shared architectural foundation into a complete dynamical ontology. Triadic Cosmos thus provides the structural and temporal layers that HST leaves implicit, while remaining fully compatible with its core principles.

Additional Architectural Distinctions. In light of the extended architectural foundations introduced in Section 2, several further distinctions between HST and the Triadic Cosmos become explicit.

First, Triadic Cosmos adopts a *global Hilbert–space ontology* in which the universe is described by a single Hilbert space with closures represented as nested finite–dimensional subspaces. HST, by contrast, employs local Hilbert spaces associated with individual causal diamonds and does not formulate a global

Hilbert space encompassing all causal domains. This global structure is essential for supporting fractal closure hierarchy, dual time, and holographic consistency across scales.

Second, Triadic Cosmos makes explicit the *Planck-scale resolution bound* implied by finite holographic capacity. Although HST incorporates horizon area as a measure of informational degrees of freedom, it does not formulate a minimal bulk resolution scale or interpret the Planck length as an information-theoretic limit on distinguishability within emergent geometry.

Third, Triadic Cosmos supplements HST with *triadic structural filters* that determine which GR geometries are physically realizable. These filters exclude configurations lacking energetic, geometric, or informational roles, violating holographic capacity, or failing triadic closure. HST does not provide such constraints, leaving the physicality of exotic GR geometries—including white holes, traversable wormholes, and Gödel-type universes—architecturally unspecified.

Together, these distinctions clarify that Triadic Cosmos extends HST not only by adding dynamical and fractal layers, but also by supplying the global, holographic, and structural constraints required for a complete architectural ontology.

3.3 Summary

The relationship between HST and the Triadic Cosmos is therefore one of architectural compatibility and conceptual extension. HST exemplifies a mainstream approach in which time is fundamental, geometry is emergent, and holography is central. Triadic Cosmos builds upon this foundation by introducing dual time, fractal closure organization, triadic recursion, vacuum architecture, and a global cosmic boundary. In this sense, Triadic Cosmos may be viewed as a dynamical completion of the holographic paradigm represented by HST.

In the next section, we broaden this perspective by examining how the Triadic Cosmos aligns with other mainstream frameworks that adopt similar holographic and emergent principles.

4 Compatibility with Mainstream Frameworks

The architectural principles underlying the Triadic Cosmos are not isolated assumptions but reflect a broader convergence within modern theoretical physics. Several mainstream frameworks adopt similar foundational ideas regarding holography, emergent geometry, fundamental time, and horizon-bounded informational structure. In this section we outline the compatibility of the Triadic Cosmos with these approaches, demonstrating that the framework provides a unified architectural completion rather than a departure from established theoretical directions.

4.1 AdS/CFT and Holographic Duality

The AdS/CFT correspondence provides the most well-developed realization of holography. Boundary time in the conformal field theory serves as the fundamental temporal parameter, while bulk geometry emerges from boundary entanglement and operator structure. This aligns directly with the Triadic Cosmos identification of electromagnetic time as fundamental and bulk geometry as a holographic decode of boundary information. Although Triadic Cosmos is not restricted to asymptotically AdS spacetimes, the architectural principles of boundary encoding and bulk emergence remain fully compatible with the duality.

4.2 Tensor-Network Emergent Geometry

Tensor-network models, including MERA and related constructions, demonstrate how spatial geometry can arise from entanglement patterns in a discrete boundary system. These models provide explicit realizations of the principle that geometry is emergent from informational structure. The Triadic Cosmos incorporates this insight at the architectural level: bulk geometry is interpreted as a decode of boundary

information, and the triadic recursion cycle provides a dynamical mechanism for the evolution of this geometry within closures.

4.3 Causal Set Theory

Causal set theory treats time as fundamental, encoded in the partial order of causal relations. Geometry is emergent from the counting measure and causal structure of the set. This matches the dual-time architecture of the Triadic Cosmos, in which electromagnetic time serves as the fundamental temporal parameter and material time emerges only within closures. The emphasis on causal structure and horizon-bounded domains provides a natural compatibility between the two frameworks.

4.4 Emergent Gravity

Emergent gravity approaches, including thermodynamic and entropic formulations, interpret gravity as an emergent phenomenon arising from informational constraints. This perspective aligns with the Triadic Cosmos view in which gravitational behavior reflects geometric responses to boundary information. The triadic recursion cycle provides a structural mechanism linking information, geometry, and energy, thereby offering an architectural completion of emergent gravity principles.

4.5 Black-Hole Thermodynamics

Black-hole thermodynamics establishes horizon area as a measure of informational capacity and links temporal evolution to mass and entropy. These principles are central to the Triadic Cosmos: black holes represent minimal closures, horizon area determines the rate of material time, and Hawking evaporation provides the empirical calibration for the dual-time structure. The compatibility between the two frameworks is therefore direct and structural.

4.6 Holographic Interference and Ensemble Projection

Recent approaches based on holographic interference and ensemble projection interpret bulk spacetime as a projection from a boundary ensemble of quantum geometries. This perspective is consistent with the Triadic Cosmos interpretation of the vacuum as an electromagnetic-time medium in which holographic decoding manifests as bulk geometry and matter. The absence of material time in the vacuum and the reliance on boundary information provide a natural alignment between the two frameworks.

4.7 Summary

Across these diverse approaches—AdS/CFT, tensor-network geometry, causal set theory, emergent gravity, black-hole thermodynamics, and holographic interference—a common architectural pattern emerges: holography is fundamental, geometry is emergent, time is nontrivial, and horizon structure encodes physical information. The Triadic Cosmos unifies and completes these principles by introducing dual time, fractal closure organization, triadic recursion, and a coherent vacuum architecture. As such, the framework stands firmly within the mainstream holographic tradition while providing a structurally complete architectural ontology.

5 Conclusion

The Triadic Cosmos framework provides a unified architectural ontology that integrates holography, emergent geometry, dual time, fractal closure organization, and a universal triadic recursion cycle. By treating electromagnetic time as fundamental and material time as an emergent, closure-specific quantity, the framework resolves longstanding tensions between vacuum dynamics, temporal behavior, and holographic projection. The identification of closures as horizon-bounded informational domains, ranging from black holes to the cosmic boundary, establishes a coherent structural hierarchy in which internal evolution is governed by the triadic cycle linking energy, geometry, and information.

The relationship between Triadic Cosmos and Holographic Space-Time (HST) illustrates how these architectural principles align with existing mainstream approaches. HST exemplifies a framework in which time is fundamental, geometry is emergent, and horizon structure encodes physical information. Triadic Cosmos extends this foundation by introducing dual time, fractal closure organization, a dynamical recursion mechanism, and a vacuum interpretation consistent with holographic decoding. In this sense, Triadic Cosmos may be viewed as a natural architectural completion of the holographic paradigm represented by HST.

Beyond HST, the compatibility of Triadic Cosmos with AdS/CFT, tensor-network emergent geometry, causal set theory, emergent gravity, and black-hole thermodynamics demonstrates that the framework does not stand in isolation. Instead, it synthesizes and unifies principles that recur across diverse theoretical directions. The holographic organization of information, the emergence of geometry from boundary structure, the central role of horizon domains, and the nontrivial behavior of time all find coherent expression within the Triadic Cosmos architecture.

Taken together, these elements position the Triadic Cosmos as a structurally complete and conceptually coherent extension of the holographic tradition. By consolidating recent insights into vacuum behavior, temporal structure, and closure dynamics, the framework offers a unified architectural foundation for understanding the emergence and evolution of physical reality.

A Comparison with Related Frameworks

This appendix provides a consolidated overview of mainstream theoretical frameworks that share architectural principles with the Triadic Cosmos. The table summarizes key points of alignment and divergence across holography, emergent geometry, temporal structure, horizon organization, and dynamical mechanisms.

Framework	Relevant Architectural Principles	Relation to Triadic Cosmos
Holographic Space–Time (HST)	Fundamental time; emergent geometry; causal diamonds; horizon area as information capacity	Strong alignment. Triadic Cosmos extends HST with dual time, fractal closures, triadic recursion, vacuum architecture, and universe as maximal closure.
AdS/CFT Correspondence	Boundary holography; emergent bulk geometry; entanglement–geometry correspondence	Compatible. Triadic Cosmos generalizes holography beyond AdS and incorporates boundary encoding as universal cosmic architecture.
Tensor–Network Geometry (MERA, etc.)	Geometry emergent from entanglement; discrete boundary encoding	Compatible. Triadic Cosmos incorporates entanglement–driven geometry and adds dynamical recursion and closure hierarchy.
Causal Set Theory	Fundamental time via causal order; emergent spacetime from discrete relations	Compatible. Triadic Cosmos matches fundamental EM–time and extends causal domains into fractal closures with emergent material time.
Emergent Gravity (Thermodynamic/Entropic)	Gravity as emergent from information; horizon thermodynamics	Compatible. Triadic Cosmos provides structural completion via triadic recursion and holographic boundary encoding.
Black–Hole Thermodynamics	Horizon area as entropy; Hawking radiation; mass–dependent temporal scaling	Direct alignment. Triadic Cosmos uses black holes as minimal closures and adopts Hawking scaling for dual–time calibration.
Holographic Interference / Ensemble Projection	Bulk as projection from boundary ensemble; vacuum as informational medium	Compatible. Triadic Cosmos interprets vacuum as EM–time medium and bulk as holographic decode, consistent with ensemble projection.

Table 1: Overview of mainstream frameworks and their architectural relation to the Triadic Cosmos.

B Axioms of the Triadic Cosmos and Alternative Recoveries

This appendix presents the core axioms of the Triadic Cosmos framework and analyzes the consequences of rejecting each axiom. For every axiom, we list the mainstream theoretical frameworks that become viable again when the axiom is broken. This provides an architectural stress–test of the Triadic Cosmos and clarifies its position within the broader landscape of holographic and emergent theories.

Axiom (Triadic Cosmos)	Role in the Architecture	If Rejected, Theories That Become Viable
Axiom 1: The Universe is Holographic	Boundary encodes all physical information; bulk is emergent decode.	GR as fundamental geometry; QFT in curved spacetime; Loop Quantum Gravity; non-holographic cosmologies; classical spacetime ontologies.
Axiom 2: Geometry is Emergent	Bulk geometry arises from boundary information and entanglement.	Classical GR as fundamental; Wheeler–DeWitt quantization; string theory as geometric background; causal set theory without holography.
Axiom 3: Electromagnetic Time is Fundamental	Universal temporal parameter; vacuum phenomena occur in t_{EM} .	Emergent time models; timeless quantum cosmology; Page–Wootters formalism; AdS/CFT with emergent bulk time; relational time theories.
Axiom 4: Material Time is Emergent	Closure-specific temporal flow generated by triadic recursion.	Single-time holographic theories (HST); canonical GR; string theory with global time; thermodynamic gravity without dual time.
Axiom 5: Closures Form a Fractal Hierarchy	Black holes = minimal closures; cosmic boundary = maximal closure.	Causal diamonds without fractal nesting (HST); AdS/CFT with fixed boundary; tensor networks without multi-scale closure structure.
Axiom 6: The Triadic Cycle $E \rightarrow G \rightarrow I \rightarrow E$ is Fundamental	Dynamical engine generating material time and closure evolution.	Static holographic frameworks; kinematical HST; AdS/CFT without dynamical emergence; entropic gravity without recursion.
Axiom 7: The Vacuum is Not a Closure	Vacuum has no horizon, no recursion, no material time; pure t_{EM} medium.	Dynamical vacuum models; string landscape; inflationary multiverse; QFT vacuum energy as physical; emergent spacetime from vacuum fluctuations.
Axiom 8: The Universe is the Maximal Closure	Cosmic boundary defines global holographic substrate.	Open cosmologies; infinite FRW models; AdS/CFT with external boundary; cyclic cosmologies; non-horizon cosmological models.
Axiom 9: String Theory is the Encoding Layer	Graviton as boundary operator; Calabi–Yau as encode substrate.	Non-string holography; tensor networks as primary encoding; spin networks; causal set encoding; emergent gravity without string theoretic structure.

Table 2: Axioms of the Triadic Cosmos and mainstream theories recovered when each axiom is rejected.

C Formal Definition of Closure and Architectural Diagram

This appendix provides a formal definition of *closure* within the Triadic Cosmos architecture and presents a diagrammatic representation of the global Hilbert–space ontology, the fractal nesting of closures, and the relationship between boundary encoding, emergent geometry, and triadic recursion.

C.1 Formal Definition of Closure

A **closure** is the fundamental structural unit of the Triadic Cosmos. It is defined as a finite–capacity, horizon–bounded domain whose energetic, geometric, and informational roles support triadic recursion and emergent material time. Formally, a closure is specified by the following properties:

C1. Hilbert–Space Embedding. A closure corresponds to a finite–dimensional Hilbert subspace

$$\mathcal{H}_{\text{closure}} \subset \mathcal{H}_{\text{universe}},$$

whose dimensionality is determined by the holographic area of its boundary,

$$\dim(\mathcal{H}_{\text{closure}}) \sim \exp\left(\frac{A}{4G\hbar}\right).$$

C2. Horizon–Bounded Informational Capacity. A closure possesses a well–defined horizon with finite informational capacity. The horizon serves as the geometric–informational interface through which boundary data encode bulk degrees of freedom.

C3. Triadic Recursion. A closure supports the triadic recursion cycle

$$E \rightarrow G \rightarrow I \rightarrow E,$$

which reorganizes energetic, geometric, and informational roles across successive states. This recursion generates emergent material time.

C4. Dual Temporal Structure. Electromagnetic time t_{EM} acts as the fundamental temporal parameter for boundary dynamics, while material time t_{mat} emerges internally through triadic recursion. The rate of material time is determined by horizon area and informational capacity.

C5. Fractal Nesting. Closures form a nested hierarchy:

$$\mathcal{H}_{\text{small}} \subset \mathcal{H}_{\text{medium}} \subset \mathcal{H}_{\text{large}} \subset \mathcal{H}_{\text{universe}},$$

reflecting the inclusion of causal domains and the scaling of informational capacity.

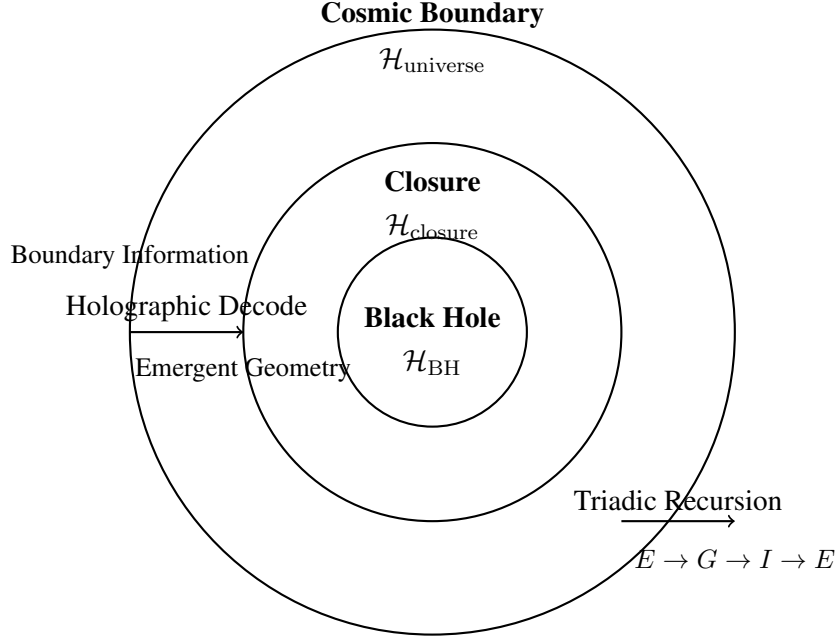
C6. Holographic Decoding. Bulk geometry, matter fields, and causal structure within a closure arise as a decode of boundary information. Geometry is emergent and continuous as a decode, while the space of physically distinguishable states is discrete due to finite boundary information.

C7. Physicality Constraints. A closure must satisfy the triadic structural filters: positivity of roles, consistent informational embedding, and full triadic closure. Geometries violating these constraints cannot correspond to physical reality.

Together, these properties define closures as the fundamental informational, geometric, and dynamical units of the Triadic Cosmos.

C.2 Architectural Diagram

The following diagram illustrates the global Hilbert–space ontology, the fractal nesting of closures, the holographic boundary, and the emergence of bulk geometry through triadic recursion. The diagram is schematic and emphasizes architectural relationships rather than metric detail.



Interpretation

The diagram shows the universe as a single global Hilbert space whose boundary encodes all physically meaningful information. Closures appear as nested finite–dimensional subspaces, each possessing its own horizon, informational capacity, and triadic recursion. Bulk geometry emerges from boundary information, and material time arises from the internal triadic cycle. Black holes represent minimal closures; the cosmic boundary represents the maximal closure.

This combined formal definition and diagram provide a compact architectural summary of the Triadic Cosmos framework.

D Black–Hole Physics in Triadic Cosmos and Holographic Space–Time

This appendix summarizes the conceptual differences between black–hole physics in the Triadic Cosmos framework and in Holographic Space–Time (HST). Although both approaches are compatible with black–hole thermodynamics, holographic entropy, and semiclassical evaporation, they differ architecturally in their interpretation of horizons, Hilbert–space structure, temporal behavior, and the role of black holes within the global organization of physical reality.

D.1 Black Holes as Minimal Closures in the Triadic Cosmos

In the Triadic Cosmos, black holes are interpreted as *minimal closures*: finite–capacity, horizon–bounded domains that instantiate the triadic recursion cycle

$$E \rightarrow G \rightarrow I \rightarrow E.$$

They correspond to finite–dimensional Hilbert subspaces,

$$\mathcal{H}_{\text{BH}} \subset \mathcal{H}_{\text{universe}},$$

whose dimensionality is determined by the holographic area of the horizon. The horizon acts as a geometric–informational interface through which boundary data encode all physically meaningful bulk degrees of freedom. The interior is not fundamental; it is an emergent decode of horizon information.

Black holes therefore represent the canonical triadic structures: they saturate energetic, geometric, and informational roles, support full triadic closure, and possess the slowest rate of emergent material time,

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2}.$$

Evaporation corresponds to the reorganization of triadic roles across successive recursion steps. In this view, a black hole is a *frozen holographic shadow* of the cosmic boundary: a minimal, self–contained closure whose internal evolution is governed entirely by triadic recursion.

D.2 Black Holes as High–Entropy States in Holographic Space–Time

In Holographic Space–Time (HST), black holes are not closures but *high–entropy states* within the Hilbert space associated with a causal diamond. Each causal diamond possesses a finite–dimensional Hilbert space whose dimension is determined by the area of its holographic screen. Black holes correspond to generic, maximally scrambled states of large causal diamonds and are characterized by fast horizon dynamics and scrambling behavior.

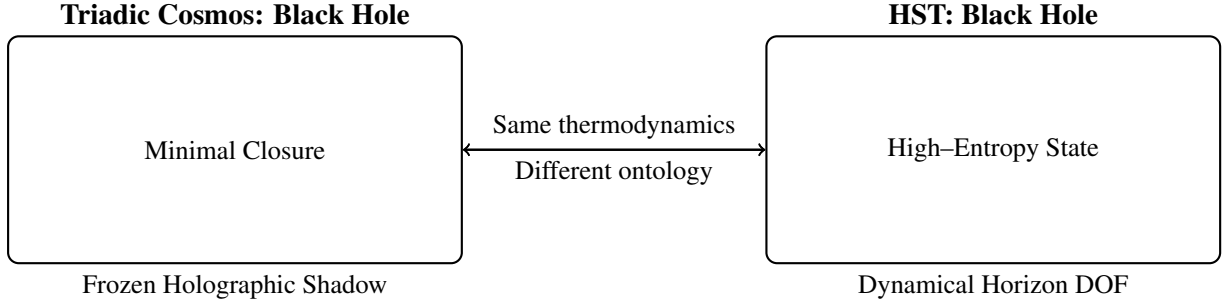
The horizon degrees of freedom in HST are dynamical: they evolve under the Hamiltonian associated with the causal diamond and determine the effective bulk geometry reconstructed from boundary data. The interior is not fundamental but is reconstructed from the time–dependent Hamiltonian flow of horizon degrees of freedom. HST employs a single fundamental time parameter and does not introduce closure–specific emergent temporal behavior.

D.3 Architectural Differences

The two interpretations differ along several structural dimensions:

- D1. Ontological Status.** In the Triadic Cosmos, a black hole is a structural unit (a closure). In HST, it is a state within a causal diamond’s Hilbert space.
- D2. Hilbert–Space Organization.** Triadic Cosmos employs a global Hilbert space with nested closure subspaces. HST employs local Hilbert spaces associated with individual causal diamonds and does not formulate a global Hilbert–space ontology.
- D3. Horizon Interpretation.** In Triadic Cosmos, the horizon is a frozen holographic shadow encoding all bulk information. In HST, the horizon degrees of freedom are dynamical and determine Hamiltonian evolution.
- D4. Temporal Structure.** Triadic Cosmos introduces dual time: fundamental electromagnetic time and emergent material time. HST employs a single fundamental time parameter and does not introduce closure–specific temporal behavior.
- D5. Interior Interpretation.** In Triadic Cosmos, the interior is a holographic decode of boundary information. In HST, the interior is reconstructed from the Hamiltonian flow of horizon degrees of freedom.
- D6. Physicality Constraints.** Triadic Cosmos imposes triadic structural filters that exclude non–physical geometries such as white holes, traversable wormholes, and Gödel–type universes. HST does not provide such filters and leaves the physicality of exotic GR geometries architecturally unspecified.

D.4 Diagrammatic Summary



Interpretation

Both frameworks reproduce the thermodynamic and holographic properties of black holes, but they differ architecturally. In the Triadic Cosmos, black holes are minimal closures with frozen holographic horizons and emergent material time. In HST, black holes are high-entropy states with dynamical horizon degrees of freedom and a single fundamental time parameter. These differences reflect the broader architectural distinctions between the two frameworks and clarify the role of black holes within the Triadic Cosmos ontology.

E Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>

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